Daksh K. Shah

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Education

New York University, Courant Institute — New York, NY
M.S. in Computer Science — *Aug. 2026 - May 2028*
Relevant Coursework: Efficient AI, Operating Systems

University of California, Santa Cruz — Santa Cruz, CA
B.S. in Computer Science; B.S. Math Theory and Computation — GPA: 3.89 — *Sep. 2022 - June 2026*
Honors/Awards: Chancellor's Award for Undergraduate Research, Dean's List (8x)
Thesis: *K-Material SDFs for Neural Attenuation Fields*
Relevant Coursework: Computer Graphics, Advanced Computer Vision (Grad), Deep Learning, Numerical Methods, Advanced Linear Algebra, Distributed Systems, Machine Learning (Grad), Artificial Intelligence (Grad)

Skills

ML Frameworks and Compute: PyTorch, CUDA, Huggingface Transformers, LoRA, WandB
ML & 3D Computer Vision: 3D Gaussian Splatting, NeRFs, Neural SDFs, Vision-Language Models, Self-supervised Learning, Representation Learning
MLOps and Infrastructure: Kubernetes, Docker, Kubeflow, DDP, Distributed Systems
Graphics and Compute Libraries: OpenCV, WebGL, OpenGL, Three.js

Publications

***K*-NeAS: Scalable Multi-Material CT Reconstruction using Neural SDFs** — ***Daksh K. Shah**, Emmanouil Nikolakakis, Razvan V. Marinescu* — Oral Presentation, Off-Grid Workshop, MICCAI (2026)
ReNAF: Regularized Neural Attenuation Fields with 3D Reconstruction Priors for Sparse-View CT — *Emmanouil Nikolakakis, **Daksh K. Shah**, Julia Wong, Julie Digne, Razvan V. Marinescu* — In Preparation

Research Experience

Research Assistant (PI: Razvan Marinescu) — January 2024 – August 2026
BiomedAI Lab, UC Santa Cruz — *Santa Cruz, CA*
- Containerized the ReNAF training pipeline in Docker and wrote parallelized Kubernetes job scripts to reserve a dedicated 8-GPU node on the NRP Nautilus cluster, enabling overnight unattended validation and supplementing a PhD student's manual runs on shared lab hardware; reran and validated metrics across 200+ experiments.

Research Assistant (PI: James Davis) — April 2025 – June 2026
Visualization and Interactive Systems (VIS) Lab, UC Santa Cruz — *Santa Cruz, CA*
- Benchmarked 3 sparse-view 3D Gaussian Splatting pipelines, including original 3DGS and Meta's official Aria splatting pipeline, across 1-2 egocentric scenes to evaluate reconstruction quality from Project Aria sensor data.
- Advanced to a cross-functional advisory role leading a weekly reading group of 5 students, covering 32 frontier papers over 6 months on Embodied Navigation, Vision-Language Models, and 3D Reconstruction.

Projects

***K*-NeAS: Scalable Multi-Material CT Reconstruction**
PyTorch, CUDA, Kubernetes — Nov. 2025 – July 2026
- Engineered a unified multi-head implicit neural representation with a shared latent backbone and K output heads, improving 3D PSNR by up to 1.8 dB over baseline NeAS as material count scales from 1 to 4, and by 1.1 dB at 10-view sparse acquisitions on clinical CBCT data.
- Designed a differentiable K-material soft selector enabling reconstruction across an arbitrary number of tissue types, where baseline methods are hard-capped at 2 materials.
- Automated attenuation bound estimation via unsupervised GMM clustering, eliminating per-scene manual hyperparameter tuning required by the baseline.

SDF-JEPA: Geometric Conditioning for 3D Vision Tasks — April 2026 – Present
- Evaluating how introducing geometric constraints on downstream finetuning on JEPA latents can recover fine details lost by JEPA pretraining.
- Diagnosed a Ceph I/O bottleneck on small files and sharded 100K files to Hugging Face datasets, leveraging faster sequential reads and DDP to reduce training cost from 400 to 160 GPU-hours.

What is the Right Moment? — October 2026 – Present
- Architected a modular multimodal pipeline using PyTorch, OpenCLIP, Silero VAD, and Whisper ASR to benchmark audio-visual keyframe selection strategies for frozen vision-language models (VLMs).

Distributed Sharded Key-Value Store | *Go, Docker* — Jan. 2025 – Mar. 2025
- Built a fault-tolerant, causally-consistent sharded key-value store in Go with vector-clock metadata, consistent hashing, and gossip-based anti-entropy; tested across 100+ containers, 12 shards, with live rebalancing.

Professional Experience

Software R&D Intern Jan. 2026 – June 2026

Keysight Technologies *Remote*

- Ported Keysight's Live Results Viewer plugin to a standalone, cross-platform executable using Avalonia, achieving full feature parity with the desktop version.
- Reverse-engineered NATS, PathWave's undocumented pub-sub messaging protocol, to research a pathway for converting Keysight's legacy desktop plugins to the web.

Leadership, Teaching, & Service

Teaching Assistant – Baskin School of Engineering, UC Santa Cruz April 2023 – June 2026

- Supported 3000+ students across 7 courses and 10 terms, including Computer Vision and Deep Learning.
- Earned Best Tutor for three consecutive terms in Computer Graphics.

Instructional Officer – Santa Cruz Artificial Intelligence Aug. 2025 – June 2026

- Delivered technical workshops on neural network architectures and training workflows to 50+ participants.
- Led a mini-hackathon and workshop on MLOps methodologies covering Kubernetes, Docker, and LoRA.